

The Programmable Body

Peptide Therapeutics, Regenerative Medicine, and Adjacent Platforms — A State-of-the-Field Whitepaper

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Executive Summary

Two decades of biology are cashing out at once. Between January 2025 and August 2026 the field crossed a threshold that is easy to miss if you track individual press releases instead of the pattern: **medicine stopped being something you take and started being something you install.**

Three shifts define the window.

1. Peptides became a platform, not a drug class. The GLP-1 story that began as diabetes control has matured into a receptor-engineering discipline. Triple-agonist retatrutide produced bariatric-surgery-magnitude weight loss in Phase 3 (28.3% at 80 weeks among completers on the top dose), oral small-molecule orforglipron broke the injection-only constraint, and a genuinely new mechanism — selective amylin agonism — delivered up to 20% weight loss in Phase 2 with a different side-effect profile. Meanwhile peptide chemistry moved into autoimmune disease, oncology conjugates, and radioligand delivery.

2. Stem cell therapy produced its first functional cures — and then attacked the reason they were impractical. Zimislecel, an allogeneic stem-cell-derived islet therapy, made 10 of 12 Type 1 diabetes patients insulin-independent at one year. That result required chronic immunosuppression, which is a poor trade for most patients. Within twelve months, hypoimmune gene-edited islets demonstrated graft survival and beta-cell function at 14 months **with no immunosuppression at all.** The bottleneck moved from "can we make the cells" to "can we hide them" — and the answer is now provisionally yes.

3. Delivery got cheap enough to skip the factory. The most consequential structural change is *in vivo* engineering. Lipid nanoparticles generated CD19 CAR T cells directly inside patients with refractory lupus — B-cell depletion and disease-activity reduction without apheresis, without a manufacturing suite, without a six-figure per-patient cell facility. In parallel, the FDA published a **Plausible Mechanism Pathway** framework (February 2026) that lets individualized gene-editing therapies be developed as a *platform* rather than one bespoke drug at a time.

The through-line: the industry's constraint has migrated from *biology* to *logistics, immunology, and regulation*. That is a much better problem to have, and it is why the next 24 months are likely to produce more approvals than the last ten years.

The counterweight: almost none of this has reached the consumer clinics selling "peptide therapy" and "stem cell treatments" today. The gap between what is FDA-approved and what is being sold in strip-mall wellness centers is now the widest it has ever been, and it is widening. Section 7 addresses that directly.

1. Scope, Method, and Evidence Tiering

This paper covers peer-reviewed results, regulatory actions, and registered trials published or announced between January 2025 and August 2026. Preprints are included only where flagged. Vendor marketing, podcast claims, and clinic websites were excluded as sources entirely.

Every claim in this document is tagged with an evidence tier. **This tiering is the single most useful thing in the paper** — most confusion in this field comes from treating Tier 4 claims as if they were Tier 1.

Tier	Definition	What it means practically
T1	FDA/EMA approved, or Phase 3 published in a peer-reviewed journal	Available or near-available; efficacy quantified
T2	Phase 1/2 results published in a peer-reviewed journal	Real signal, real humans, unproven at scale
T3	Preclinical, animal, or preprint	Directionally interesting; ~90% will not translate as described
T4	Marketed, sold, or promoted without controlled human efficacy data	Commerce, not medicine

A recurring failure mode in this sector: a T3 mouse result becomes a T4 product within eighteen months, entirely skipping T2 and T1. Compounding pharmacies and wellness clinics are structurally incentivized to make that jump. Treat any therapy you can buy without a trial enrollment as T4 until proven otherwise.

2. Part I — Peptide Therapeutics

2.1 The incretin platform matures from single agonist to receptor orchestration

The defining insight of 2025–2026 is that metabolic disease responds to *combinations* of gut-hormone receptors, and that each added receptor buys additional efficacy at the cost of tolerability.

Retatrutide (GLP-1 / GIP / glucagon triple agonist) — T1. Phase 3 TRIUMPH data reported in 2026 showed that participants without diabetes on the highest dose who remained on treatment lost an average of **28.3% of body weight at 80 weeks**. This is bariatric-surgery territory achieved pharmacologically. The critical caveat that most coverage buried: **11% of patients on the top dose discontinued due to adverse events**, and the all-participants (treatment-policy) efficacy estimate falls to ~25%. In a parallel 68-week trial in obesity with knee osteoarthritis, completers on the top dose lost 28.7% versus 23.7% across all treated participants. Notably, some participants discontinued for *perceived excessive weight loss* — a genuinely new problem in obesity pharmacology, and a preview of the body-composition debate below.

Orforglipron (oral, non-peptide small-molecule GLP-1 receptor agonist) — T1. The strategic significance of orforglipron is not its efficacy — ATAIN-1 reported ~11.2% mean weight loss over 72 weeks, with ≥10% loss in 54.6% of participants, which is below injectable tirzepatide. Its significance is **manufacturing and access**. Peptides require cold chain, injection devices, and constrained peptide-

synthesis capacity. A daily oral small molecule does not. ATTAIN-2 confirmed superiority over placebo in adults with obesity and Type 2 diabetes across 136 sites in ten countries, and ATTAIN-MAINTAIN showed orforglipron maintained **74.7% of prior weight reduction versus 49.2% on placebo** in patients stepping down from tirzepatide or semaglutide. That maintenance role — cheap oral agent holding the line after an expensive injectable does the work — is likely where the largest market sits.

CagriSema (cagrilintide + semaglutide) — T1. The REDEFINE program tested co-administration of an amylin analog with semaglutide in adults with obesity or overweight plus complications, establishing combination amylin-incretin therapy as a validated commercial strategy rather than an academic curiosity.

2.2 Amylin: the first genuinely new metabolic mechanism since GLP-1

If one result in this whitepaper deserves disproportionate attention, it is this one.

Eloralintide (selective amylin receptor agonist) — T2, strong. A 48-week Phase 2 trial (n=263, enrolled February 2024–August 2025) reported mean body-weight change of **–9% (1 mg), –12% (3 mg), –18% (6 mg), and –20% (9 mg)** versus –0.4% on placebo. Twenty percent weight loss from a Phase 2 monotherapy is remarkable. Tolerability was dose-dependent and non-trivial — nausea ranged from 11% at 1 mg to 64% at 6 mg, fatigue from 0% to 46%.

Why this matters more than the numbers: amylin signals satiety through a different central pathway than GLP-1. That opens three doors simultaneously — (a) patients who cannot tolerate incretins get an alternative; (b) combination therapy can hit two mechanisms at sub-maximal doses of each, potentially decoupling efficacy from side effects; (c) preclinical and early clinical data suggest amylin agonists may better preserve lean mass, which is the single most important unresolved question in the entire obesity-drug category.

Petrelintide — T2. Zealand Pharma's long-acting amylin analog advanced into ZUPREME-1 and ZUPREME-2 Phase 2b dose-finding programs (n≈480), covering obesity and obesity with Type 2 diabetes.

The open question no one has answered: at 25–30% total weight loss, what fraction is fat and what fraction is skeletal muscle and bone? DEXA sub-studies remain underpowered and inconsistently reported. Until that is settled, the long-term functional outcome of these drugs in patients over 60 is genuinely unknown. Anyone claiming certainty in either direction is ahead of the data.

2.3 Peptides beyond metabolism

The metabolic story dominates headlines, but the pipeline broadened materially. Over 150 peptides are in active clinical trials with several hundred more in preclinical development, spanning oncology, neurodegeneration, infectious disease, autoimmune conditions, and pain.

Peptide-drug conjugates and radioligand therapy — T1/T2. The PDC field has ~96 candidates in development with roughly six in Phase 3, though the approved landscape remains thin: Lutathera (a somatostatin-agonist/DOTA/¹⁷⁷Lu construct for neuroendocrine tumors) is the standing proof of concept after Pepaxto's US withdrawal. Radioligand therapy has meaningfully reshaped advanced prostate cancer treatment by pairing molecular imaging with targeted radionuclide delivery — a genuine "see it, then hit it" precision loop.

Autoimmune peptides — T2, treat with caution. Reports of oral and subcutaneous peptide agents

producing clinically meaningful improvement in systemic lupus erythematosus circulated during this window. Several of these claims trace to secondary sources rather than primary journal publication. **We flag this area as under-verified** and recommend confirming any specific autoimmune peptide claim against the primary trial registration and publication before acting on it.

2.4 The design revolution: computational peptide and minibinder engineering

This is the part of the field with the longest tail and the least public attention.

De novo protein and peptide design — T2/T3, accelerating. The mature workflow is now standardized: RFdiffusion generates backbones, ProteinMPNN optimizes sequences, AlphaFold2/3 screens for structural plausibility. Deep learning methods now design high-affinity protein-binding **macrocycles** — a peptide-adjacent modality that combines antibody-like specificity with small-molecule pharmacokinetics.

The therapeutically important property is size. Recent work reports that designed **minibinders distribute at higher concentrations within subcutaneous and intracranial tumors and cross the blood–brain barrier more efficiently than antibodies**. An orally-dosed IL-23R-directed minibinder reportedly outperformed clinical antibodies on GI stability, pharmacokinetics, and efficacy — at lower production cost. If that result generalizes, it threatens the economics of a substantial fraction of the monoclonal antibody market.

Honest counterweight — T3. Independent evaluations have found RFdiffusion success rates for *functional* binder design (as opposed to binding-competent design) to be low in some applications. The gap between "binds the target" and "does the therapeutic job" remains wide. Generative design has compressed the discovery timeline dramatically; it has not eliminated the attrition.

AI-discovered small molecules — T2, milestone. Rentosertib (formerly ISM001-055), a first-in-class TNIK inhibitor for idiopathic pulmonary fibrosis where both the *target* and the *molecule* were identified by generative AI, completed a randomized, double-blind, placebo-controlled Phase 2a trial (n=71 across three dose arms plus placebo), published in *Nature Medicine* in June 2025. Very few AI-originated drugs have reached human trials at all; this is the reference case, and IPF is a disease with no therapy that reverses the degenerative course.

3. Part II — Stem Cell and Regenerative Medicine

3.1 The functional cure arrives: stem-cell-derived islets in Type 1 diabetes

Zimislecel (allogeneic stem-cell-derived, fully differentiated islet cells) — T1. Published in the *New England Journal of Medicine*, the FORWARD trial reported that **10 of 12 participants (83%) were insulin-independent at day 365**. All participants showed engraftment and islet function via detectable C-peptide. The primary endpoint in Part C was freedom from severe hypoglycemic events between days 90 and 365 with HbA1c below 7% or a ≥ 1 percentage-point reduction from baseline.

This is, without hedging, a functional cure for Type 1 diabetes in the treated population.

The catch, stated plainly: all participants received glucocorticoid-free chronic immunosuppression. For a disease that is well-managed by many patients with modern closed-loop insulin pumps, trading

autoimmunity for lifelong immunosuppression and its infection and malignancy risks is a defensible choice only for patients with severe hypoglycemia unawareness. The therapy as constituted is not for the average Type 1 patient.

3.2 Hypoimmune engineering: removing the reason cell therapy was impractical

Hypoimmune (HIP) allogeneic islets — T2, field-defining. The hypoimmune platform genetically ablates HLA class I and class II expression and upregulates CD47, rendering transplanted cells effectively invisible to both adaptive and innate immune surveillance. In a Type 1 diabetes patient who received genetically modified allogeneic islets **with no immunosuppression whatsoever**, results reported in *NEJM* on July 10, 2026 documented sustained beta-cell function and absence of an immune response against the allograft through 14 months.

Supporting evidence spans humanized mouse models, cynomolgus macaques, and fully immunocompetent allogeneic rhesus macaques, where hypoimmune iPSCs survived long-term.

Why this is the most important item in the whitepaper: immunosuppression is the tax that made every allogeneic cell therapy a bad deal for anything short of a life-threatening indication. If hypoimmune engineering holds at scale, that tax goes away — and every cell therapy indication gets re-underwritten. Islets, dopaminergic neurons, cardiomyocytes, retinal pigment epithelium, cartilage. The bull case for regenerative medicine over the next decade rests substantially on whether $n=1$ at 14 months becomes $n=100$ at 5 years.

The honest read: it is one patient. Durability, tumorigenicity surveillance in cells engineered to evade immune clearance (a genuine and non-theoretical concern), and manufacturing consistency are all unresolved.

3.3 Neurological cell replacement: real progress, no finish line yet

Bemdaneprocel (hESC-derived dopaminergic neuron progenitors, Parkinson's disease) — T2. The Phase 1 trial transplanted bilateral grafts into the putamen of twelve patients, with **no serious adverse events at 24 months**. Additional Phase 1 and Phase 1/2 open-label hESC-derived dopaminergic cell results were published in *Nature* (2025) and *Nature Medicine* (2026).

Status as of August 2026: the Phase 3 trial (NCT06944522, "exPDite-2") began recruiting June 17, 2025 with estimated primary completion in **March 2027**. **Efficacy is not established.** Any source claiming a Phase 3 Parkinson's cell-therapy result today is wrong.

Adjacent — mesenchymal stem cells in Parkinson's — T2. A Phase IIa randomized placebo-controlled trial of MSCs as disease-modifying therapy in idiopathic Parkinson's (NCT04506073) is ongoing. MSC results across indications have historically underdelivered relative to enthusiasm; weight accordingly.

3.4 Transplant immunology: the year's clearest regulatory win

Tregzi (allogeneic Treg-based immunotherapy with HSPCs and T cells) — T1, FDA approved June 30, 2026. Developed by Orca Bio and approved for use in matched-donor hematopoietic stem cell transplantation in adults with hematologic malignancies, Tregzi is the first regulatory-T-cell-based immunotherapy approved to improve chronic-GVHD-free survival.

The product is three sequentially administered components sourced from mobilized peripheral blood of an 8/8 HLA-matched related or unrelated donor: purified hematopoietic stem and progenitor cells, regulatory T cells, and conventional T cells.

PRECISION-T efficacy (NCT05316701): 187 patients randomized to Tregzi plus single-agent tacrolimus prophylaxis (n=93) versus unmanipulated allograft plus tacrolimus and methotrexate (n=94). Median chronic-GVHD-free survival was **not estimable in the Tregzi arm versus 7.3 months in the control arm (HR 0.26, 95% CI 0.14–0.47, p<0.00001)**.

A hazard ratio of 0.26 is an unusually large effect. The mechanistic lesson generalizes beyond transplant: **precisely composed cell products beat bulk cell products**. Controlling the ratio of regulatory to effector therapy.

3.5 The unglamorous bottleneck: manufacturing, cost, and access

None of the above scales on current infrastructure. Autologous and precisely-composed allogeneic cell products require GMP suites, cryopreservation logistics, HLA matching infrastructure, and per-patient release testing. The cost structure resembles organ transplantation more than pharmaceuticals.

Three responses are visible in the data: off-the-shelf allogeneic products (bemdaneprol is cryopreserved and off-the-shelf), hypoimmune engineering to eliminate matching and immunosuppression, and — most disruptively — *in vivo* engineering that eliminates ex vivo manufacturing entirely. See Section 4.1.

4. Part III — Adjacent Platforms Reshaping Both Fields

4.1 In vivo CAR-T: the manufacturing problem solved by not manufacturing

LNP-delivered in vivo CD19 CAR-T in refractory SLE — T2, structurally important. In patients with refractory systemic lupus erythematosus, lipid nanoparticles delivering CAR-encoding mRNA generated CD19 CAR T cells *inside the patient*, producing **B-cell depletion, reduced disease activity, and no major toxic effects**. Early results in five patients were published in the *New England Journal of Medicine*; the underlying platform work appeared in *Science*.

In non-human primate work, B cells that reconstituted after depletion were **predominantly naïve** — consistent with an immune-system reset rather than mere suppression. That is the mechanistic basis for the hypothesis that autoimmune disease may be *resettable* rather than permanently managed.

Why this reorders the industry: conventional CAR-T requires apheresis, ex vivo transduction, expansion, lymphodepleting chemotherapy, and a specialized center — weeks of turnaround and a cost measured in hundreds of thousands of dollars. An LNP infusion is a vial. If in vivo CAR-T holds up, cell therapy stops being a hospital procedure and becomes a drug. That single change is worth more to patient access than any efficacy improvement on this list.

Related work on CD7-targeted LNPs for mRNA delivery to T cells (preprint, T3) and generative-AI-designed minibinder CAR domains (T3) suggests the targeting layer will improve quickly.

4.2 Personalized gene editing becomes a regulated platform

The most consequential regulatory development of the window received almost no mainstream coverage.

FDA Plausible Mechanism Pathway — February 2026 draft guidance, T1 (regulatory). The FDA published a framework for developing and approving **individualized therapies for genetic conditions**. Alongside it, the agency issued draft guidance (June 2, 2026) to accelerate gene therapies by making greater use of existing scientific and regulatory knowledge — explicitly enabling platform-based rather than product-by-product review.

The precedent case — T2. An individualized base-editing therapy was administered to an infant with

access IND. Through formal FDA interaction, a path was outlined for a **customizable in vivo base-editing platform covering seven infantile-onset urea cycle disorders plus phenylketonuria**. Children's Hospital of Philadelphia planned an umbrella trial for urea cycle disease starting in 2026.

Why this is a bigger deal than any single drug: ultra-rare genetic disease has always been economically unservable — you cannot run a pivotal trial on four patients worldwide, and no sponsor can amortize a program across them. A platform pathway means the *editing machinery* is validated once and the *guide RNA* becomes the variable. That is the difference between a bespoke suit and a tailored one, and it converts thousands of "untreatable" mutations into an engineering exercise.

Also T1-adjacent: Intellia advanced the first *in vivo* CRISPR-based gene editing therapy toward FDA filing, reported in *Nature Biotechnology* in 2026.

4.3 Xenotransplantation exits the anecdote phase

EXPAND (United Therapeutics UKidney) — T2, first regulated clinical program. A multicenter, open-label interventional trial of a **10-gene-edited porcine kidney** — six human gene insertions for immunologic compatibility, four porcine knockouts to reduce rejection and control graft growth — in end-stage renal disease patients ineligible for or unlikely to receive a human kidney within five years. The first transplant under protocol was performed at NYU Langone on **November 3, 2025**.

Benchmark: a 67-year-old recipient at Massachusetts General Hospital reached **271 days** — the longest reported pig-to-human kidney survival — before graft removal for progressive dysfunction in October 2025.

271 days is not a solution to organ scarcity. It is proof that the timeline is now measured in months rather than days, which is what a clinical program needs to iterate against.

4.4 Neuromodulation: where movement-disorder gains are actually coming from

For neurological movement disorders specifically — including dystonia — it is important to be blunt: **neither peptide therapeutics nor stem cell replacement has produced a treatment for dystonia in this window**. The cell-replacement work is dopaminergic-neuron-specific and Parkinson's-targeted. Extrapolating it to dystonia is not supported.

The real near-term progress is in **device and stimulation engineering**:

- **Adaptive (closed-loop) DBS — T2/T1.** aDBS senses neural biomarkers and modulates stimulation in real time rather than delivering continuous fixed-parameter current. Dual-threshold aDBS mode is

implemented across the Medtronic Percept device family, with chronic clinical use reported in Parkinson's cohorts enrolled January–April 2025 at the Parkinson Institute of Milan.

- **Directional leads and remote programming — T1.** Steerable current fields plus telemedicine-enabled programming materially reduce stimulation-induced side effects and eliminate travel burden for tuning visits.
- **MR-guided focused ultrasound — T1/T2.** Incisionless lesioning is an established option in tremor and is under active study in broader parkinsonism and related disorders.
- **Trial to watch:** ADVENT, a prospective multicenter randomized double-blind crossover trial comparing adaptive versus conventional STN-DBS in levodopa-responsive Parkinson's, with an estimated start of **August 2026**.

Field consensus work (Delphi studies in *npj Parkinson's Disease*) suggests aDBS is transitioning from research to real clinical option. DBS is already a standard therapy for selected dystonia patients; the adaptive and directional upgrades are the realistic source of improvement over the next three to five years — not regenerative medicine.

5. Cross-Cutting Theses

Thesis 1 — The binding constraint moved from discovery to delivery. We can now design a binder for nearly any target. Getting it to the right cell, at the right dose, without immune clearance, at a manufacturable cost is the entire remaining problem. Capital and talent should be read accordingly: LNP chemistry, tissue targeting, and immune evasion are where the leverage is.

Thesis 2 — Immune evasion is the master key to regenerative medicine. Hypoimmune engineering does not improve one therapy; it re-prices the entire allogeneic cell therapy sector by deleting the immunosuppression tax. Watch this one variable above all others.

Thesis 3 — "Reset" is displacing "manage" in autoimmune disease. Naïve B-cell reconstitution after in vivo CAR-T depletion implies a durable immunological reset. If confirmed, the chronic-biologic business model for lupus, MS, and rheumatoid arthritis faces an existential challenge from one-time therapies.

Thesis 4 — Regulation became a technology. The FDA's plausible-mechanism and platform-knowledge frameworks are, functionally, an innovation in the same sense as a new vector. They change what is economically possible to develop. Under-covered and under-appreciated.

Thesis 5 — AI compressed discovery, not attrition. Rentosertib and de novo minibinders prove generative methods produce clinic-ready candidates faster. Nothing yet shows they produce candidates that *fail less often* in Phase 2 and 3, which is where 90% of the cost lives. Claims that AI will collapse total drug development cost remain unproven.

Thesis 6 — Composition beats quantity in cell therapy. Tregzi's HR of 0.26 came from controlling the *ratio* of cell subtypes, not from more cells. Expect precise cell composition to become a standard design axis across the sector.

Thesis 7 — Tolerability, not efficacy, is now the competitive battleground in metabolic disease. When multiple agents deliver 20–28% weight loss, the winner is whoever gets there with the fewest discontinuations and the best preserved lean mass. Retatrutide's 11% top-dose discontinuation rate is the

number to beat.

6. Risk Register

Risk	Field	Assessment
Body-composition loss (lean mass, bone) at 25–30% total weight reduction	Peptides	Under-characterized. Highest-consequence open question in metabolic medicine
Tumorigenicity in immune-evasive engineered cells	Regenerative	Structural concern — cells designed to evade surveillance also evade cancer surveillance
Durability beyond 24 months for cell grafts	Regenerative	Unknown; longest hypoimmune human data is 14 months, n=1
Off-target and bystander editing in personalized base editing	Gene editing	Managed but not eliminated; bespoke PAM-variant editors are the current mitigation
Zoonotic transmission	Xenotransplant	Explicitly monitored in EXPAND; no clinical events reported to date
Manufacturing cost and access inequity	All	The default outcome is that these therapies serve the insured wealthy first
Unregulated clinic market outrunning the evidence	All	Active harm occurring now. See Section 7
Long-term LNP re-dosing immunogenicity	In vivo CAR-T	Largely unstudied; repeat dosing may be limited

7. The Evidence Gap: What Is Sold vs. What Is Proven

This section exists because the commercial market in "peptide therapy" and "stem cell treatment" bears almost no relationship to the science above.

The FDA maintains standing consumer alerts on regenerative medicine products including stem cells and exosomes. These are not advisory footnotes — they exist because patients have been harmed by unapproved products, including infections and vision loss.

Practical filters for evaluating any clinic or product:

- 1. Ask for the trial registration number.** NCT-numbered, publicly registered, with published results. If the answer is a testimonial or a podcast, it is T4.
- 2. "FDA-registered facility" ≠ "FDA-approved therapy."** Facility registration is an address filing. Approval is a data package. Marketers conflate these deliberately.
- 3. Approved stem-cell therapies are a short list.** Hematopoietic stem cell transplantation for blood cancers and defined disorders, a small set of cord-blood products, and now Tregzi. Autologous fat- or PRP-derived "stem cell" injections for joints, energy, or anti-aging are not on it.

4. **Compounded and research-only peptides.** Substances sold "for research use only" with dosing advice attached are being sold outside the regulatory system on purpose. Sterility, identity, and purity are unverified.
5. **The one-way door test.** Peptides clear. Injected cells and gene edits do not. Match the reversibility of the intervention to the strength of the evidence behind it.
6. **Legitimate access to frontier therapy is free.** It is called clinical trial enrollment. Search clinicaltrials.gov. If a frontier therapy has a price tag and no protocol, that is the tell.

8. Watch List: Catalysts, Late 2026 – 2028

Catalyst	Field	Expected	Why it matters
exPDite-2 (bemdaneprocet) Phase 3 primary completion	Regenerative	March 2027	First controlled efficacy read for cell replacement in Parkinson's
Hypoimmune islet cohort expansion beyond n=1	Regenerative	2026–2027	Determines whether immunosuppression-free cell therapy generalizes
CHOP urea-cycle umbrella trial	Gene editing	From 2026	First real-world test of the platform regulatory pathway
ADVENT adaptive vs. conventional DBS	Neuromodulation	Start Aug 2026	Definitive evidence for closed-loop stimulation
Retatrutide regulatory decisions	Peptides	2026–2027	Sets the tolerability/efficacy frontier for the category
Amylin Phase 3 programs (petrelintide, eloralintide)	Peptides	2027	Whether the lean-mass-preservation hypothesis survives contact with Phase 3
In vivo CAR-T expansion cohorts in autoimmune disease	Cell therapy	2026–2027	Whether the immune "reset" is durable
EXPAND xenotransplant multi-patient survival data	Xenotransplant	2026–2028	Whether 271 days becomes years
First minibinder therapeutic IND	Computational design	2026–2028	Whether AI-designed proteins reach the clinic as products
Orforglipron launch economics	Peptides	2026–2027	Whether oral incretins actually expand access or just extend patents

9. Practical Implications

For patients and individuals. The highest-value action available today is not purchasing anything — it is knowing which registry to watch and which trials are recruiting for your specific condition. For movement disorders specifically, the near-term gains are in adaptive and directional neurostimulation, available through established neurosurgical centers, not in regenerative products sold direct-to-consumer.

For operators and investors. The asymmetry sits in delivery and immune evasion, not in target discovery. Any thesis premised on "we found a novel target" is competing against generative models that find targets cheaply. Any thesis premised on "we can get this payload into that tissue without immune clearance, at cost" is competing against a much shorter list.

For technologists. The bottlenecks are now data and logistics problems: manufacturing consistency, cold-chain and release testing orchestration, trial matching, real-world outcome capture, and regulatory documentation under platform pathways. These are ordinary software problems attached to extraordinary biology, and they are systematically underserved.

10. Conclusion

The 2025–2026 window will be read as the point at which regenerative medicine stopped being a promise and became an engineering discipline with a cost curve. Stem-cell-derived islets cured Type 1 diabetes in most of a small cohort. Gene editing hid transplanted cells from the immune system for over a year without a single immunosuppressant. Lipid nanoparticles built CAR T cells inside a living patient. A regulator wrote a pathway for medicines that are manufactured for one person. And peptide chemistry, the oldest modality on this list, produced weight loss that rivals surgery.

None of it is finished. Durability is unproven, costs are unacceptable, manufacturing is fragile, and the consumer market is running years ahead of the evidence in ways that are actively hurting people.

But the direction is unambiguous. The body is becoming programmable — and the remaining obstacles are the kind engineers solve, not the kind biologists discover.

Appendix A — Primary Sources

Peptide therapeutics

- [Orforglipron, an Oral Small-Molecule GLP-1 Receptor Agonist, in Early Type 2 Diabetes — NEJM](#)
- [Orforglipron for obesity in people with type 2 diabetes \(ATTAIN-2\) — The Lancet](#)
- [ATTAIN-MAINTAIN phase 3b trial — Nature Medicine](#)
- [Eloralintide phase 2, 48-week trial — The Lancet](#)
- [Eloralintide: from discovery to clinical proof of concept — PMC](#)
- [Triple-Hormone-Receptor Agonist Retatrutide for Obesity, Phase 2 — NEJM](#)
- [Retatrutide TRANSCEND-T2D-1 phase 3 — The Lancet](#)
- [Lilly's 'triple-G' drug leads to bariatric-surgery levels of weight loss — STAT](#)
- [Coadministered Cagrilintide and Semaglutide — NEJM](#)
- [TRIUMPH registrational trial design — PubMed](#)
- [Peptide-drug conjugates in tumor therapy — PubMed](#)
- [Current and future perspectives on radioligand therapy in advanced prostate cancer — PMC](#)

Stem cell and regenerative medicine

- [Stem Cell–Derived, Fully Differentiated Islets for Type 1 Diabetes \(zimislecel\) — NEJM · PubMed](#)
- [Long-Term Survival of Hypoimmune Allogeneic Islets without Immunosuppression — NEJM](#)
- [Survival of Transplanted Allogeneic Beta Cells with No Immunosuppression — NEJM](#)
- [Human hypoimmune primary pancreatic islets avoid rejection — Science Translational Medicine](#)
- [Hypoimmune iPSCs survive long term in immunocompetent allogeneic rhesus macaques — Nature Biotechnology](#)
- [Phase I trial of hES cell-derived dopaminergic neurons for Parkinson's disease — Nature](#)
- [hESC-derived dopaminergic cells for Parkinson's, phase 1/2 — Nature Medicine](#)
- [Gene and cell therapies for Parkinson's make headway — Nature Biotechnology](#)
- [exPDite-2 bembdanepocel Phase 3 — ClinicalTrials.gov NCT06944522](#)
- [FDA approves Tregzi \(allogeneic Treg immunotherapy with HSPC and T cells-vldq\)](#)
- [FDA press announcement — donor immune cells to prevent GVHD](#)
- [TREGZI product page — FDA](#)
- [FDA approves Orca Bio's T cell therapy for blood cancer patients — STAT](#)

Adjacent platforms

- [In Vivo CD19 CAR T-Cell Therapy for Refractory Systemic Lupus Erythematosus — NEJM](#)
- [In vivo CAR T cell generation to treat cancer and autoimmune disease — Science](#)
- [In vivo CAR-T cell engineering in refractory SLE — Nature Reviews Rheumatology](#)
- [FDA's New Plausible Mechanism Pathway — NEJM](#)
- [Implications of the FDA's plausible mechanism framework for personalized in vivo prime editing — PMC](#)
- [FDA Issues Draft Guidance to Help Accelerate Cell and Gene Therapies](#)
- [Celebrating a new, faster path to gene-editing medicines on demand — STAT](#)
- [Intellia heads to FDA with first in vivo CRISPR-based gene editing therapy — Nature Biotechnology](#)
- [The First Clinical Renal Xenotransplantation Study \(EXPAND\) — PMC](#)
- [Consistent survival in life-supporting porcine kidney xenotransplantation with 10GE source pigs — Nature Communications](#)

Computational design and AI

- [The past, present and future of de novo protein design — Nature](#)
- [Accurate de novo design of high-affinity protein-binding macrocycles — Nature Chemical Biology](#)
- [De novo protein design: a transformative frontier in clinical protein applications — PMC](#)
- [RFDiffusion low success rate in functional binder design — bioRxiv \(preprint\)](#)
- [A generative AI-discovered TNIK inhibitor for IPF: randomized phase 2a — Nature Medicine](#)

Neuromodulation

- [From adaptive deep brain stimulation to adaptive circuit targeting — Nature Reviews Neurology](#)
- [Will adaptive DBS for Parkinson's become a real option soon? Delphi consensus — npj Parkinson's Disease](#)

- [Advances in deep brain stimulation for movement disorders — PubMed](#)
- [Deep brain stimulation for dystonia — PMC](#)

Consumer protection

- [FDA Consumer Alert on Regenerative Medicine Products Including Stem Cells and Exosomes](#)
- [Important Patient and Consumer Information About Regenerative Medicine Therapies — FDA](#)

Appendix B — Verification Notes

Claims in this document were compiled from the primary sources listed in Appendix A. Two categories carry explicit caveats.

1. **Autoimmune peptide results (Section 2.3)** were encountered primarily via secondary aggregators rather than confirmed primary publications. They are flagged in-text and should be independently verified before citation.
2. **Direct full-text verification of NEJM and FDA pages was blocked** by publisher access controls during compilation. Figures reported here (zimislecel 10/12; Tregzi HR 0.26; hypimmune islets at 14 months; retatrutide 28.3%) derive from indexed abstracts, FDA summaries, and journalism from STAT News. They are consistent across independent sources but should be checked against full text before use in any regulated, clinical, or investment-decision context.

No figure in this paper was estimated, interpolated, or inferred. Where a number was unavailable, it is described as unavailable.

Disclaimer

This whitepaper is an educational and analytical summary of published research. **It is not medical advice.** Nothing here should be used to start, stop, or modify any treatment. Therapies discussed range from FDA-approved to early experimental; most are not available outside clinical trials. Consult a licensed physician regarding any medical decision. The author has no financial interest in any company, product, or therapy named.

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